BASICS OF INFECTION & IMMUNITY Module (BioMM - 2152)
for Anesthesia year II students

Introduction to Medical Microbiology

Kelemework Adane (PhD)
Dpt of Microbiology, Immunology & Parasitology CHS, AAU

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Learning objectives

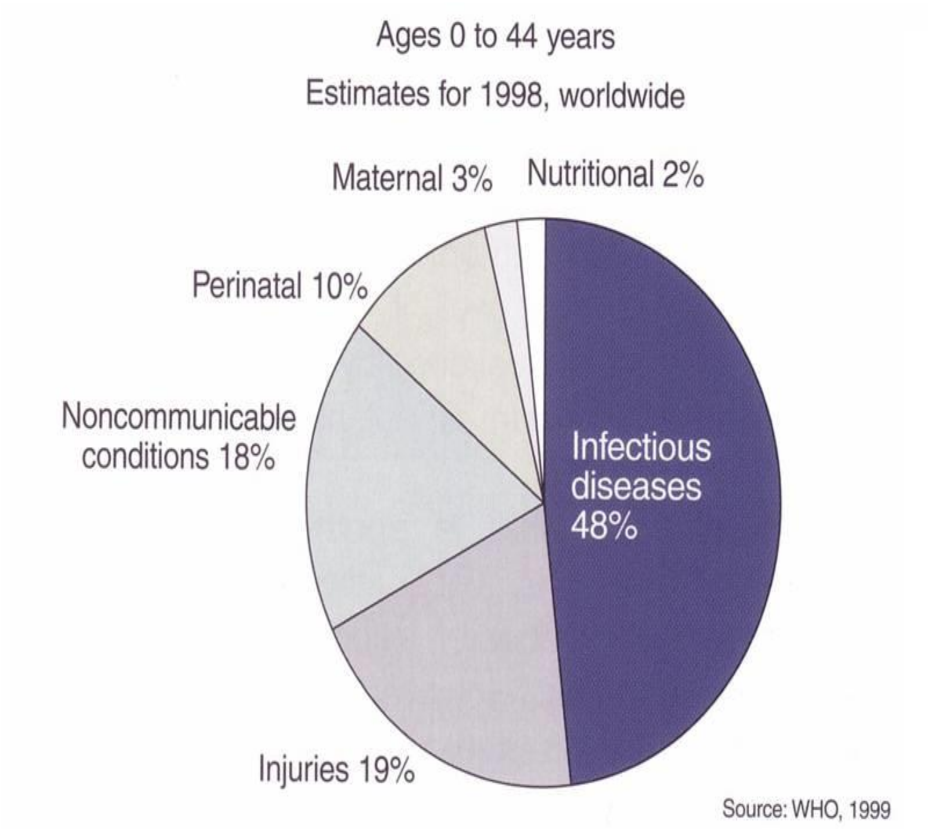
Upon completion of this lesson you are expected to:

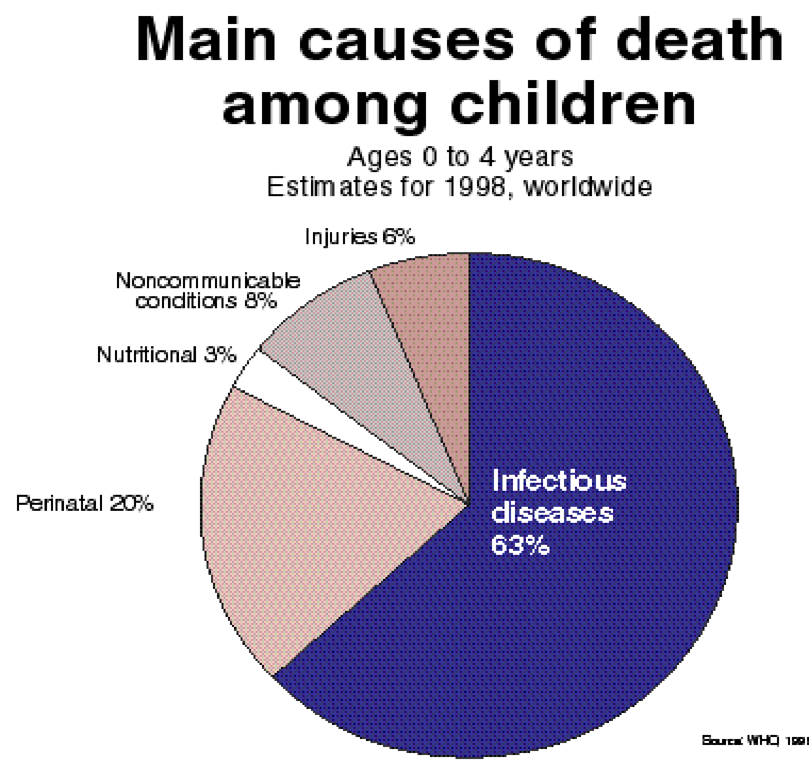
- Define microbiology and microorganisms
- Explain how microorganisms affect our lives
- Describe the history of microbiology and germ theory of diseases
- Explain the basis for classifying and naming of microorganisms
- Describe eukaryotic and prokaryotic cells

Why you need to study about Microbiology?

Infectious disease : leading causes of death

53.9 million from all causes, worldwide





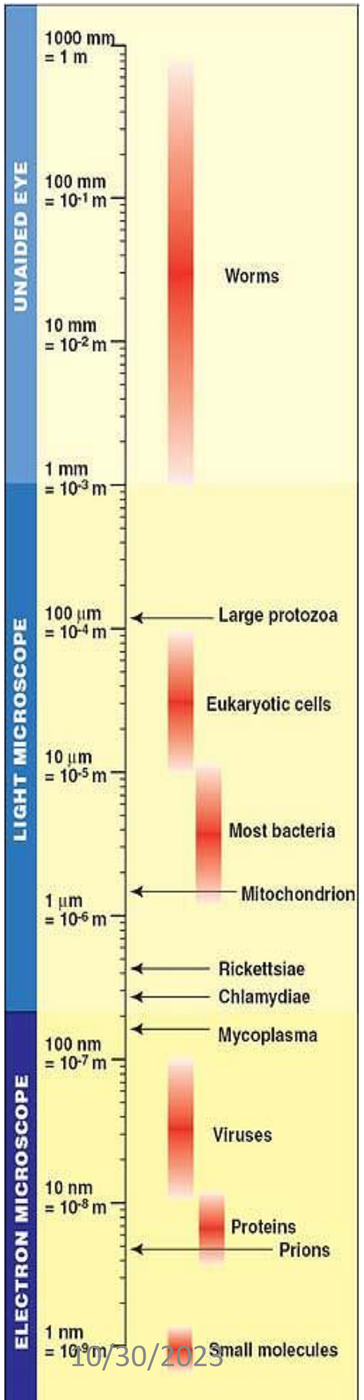
What is Microbiology?

- The study of organisms too small to be seen by the naked eye- **microorganisms**
- The word microbiology is derived from three Greek words:

  Micros – small

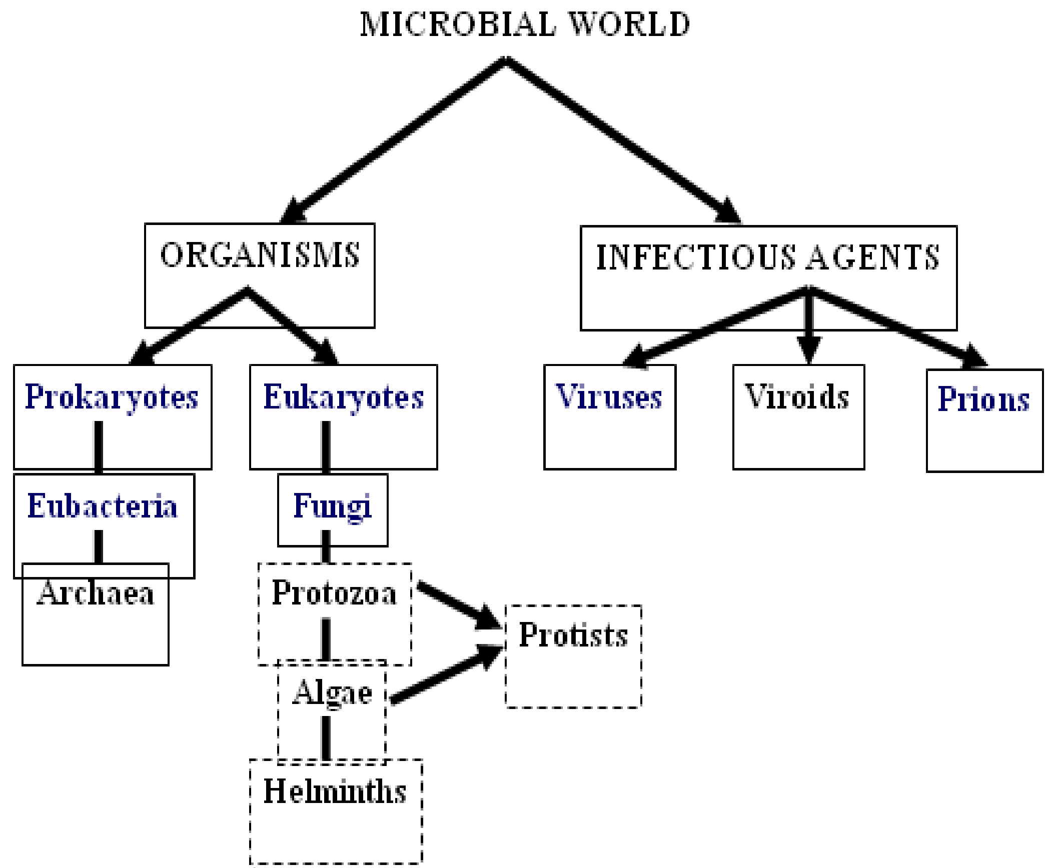
  Bios – life

  Logos - science



- **What are microorganisms ?**
- too small to be seen with the naked eye, very diverse and found almost everywhere

Relative size of organisms and molecules



Medical microbiology

- The study of **pathogenic microbes** and their role in human illness

- Includes

  - **Bacteriology:** study of prokaryotes (single-celled)
  - **Virology**: Study of non-cellular, parasitic infectious agents
  - **Mycology**: Study of Fungi; Microscopic (molds & yeasts)

    : Macroscopic (mushrooms)

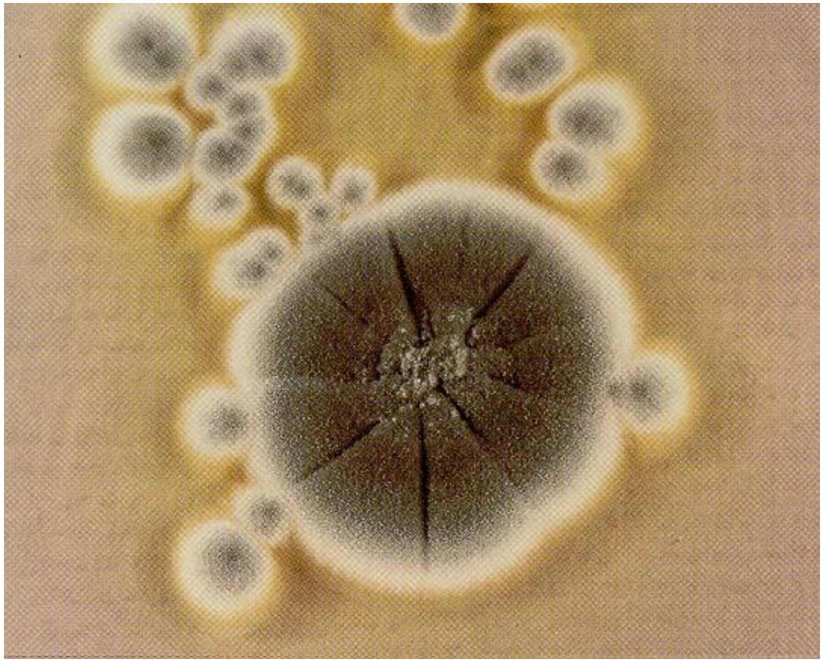
Beneficial & harmful aspects microorganisms

Beneficial aspects

- Recycles vital elements (carbon, nitrogen, sulfur, hydrogen, and oxygen)
  - Maintain balance of environment
- Nitrogen fixation
- Photosynthesis
- Recycles water (sewage treatment)
- Clean up toxic wastes (bioremediation)
- Manufacture of food and drink

Beneficial aspects…

Microbes are used to produce Antibiotics



- 1929 Alexander Fleming developed Penicillin from mold
  - *Penicillium notatum*

Beneficial aspects…

- **Bacteria synthesize chemicals that our body needs**
- Example: *Escherichia coli*
  - B vitamins - for metabolism
  - Vitamin K - blood clotting

Harmful Effects

- Cause disease & basis for bioterrorism

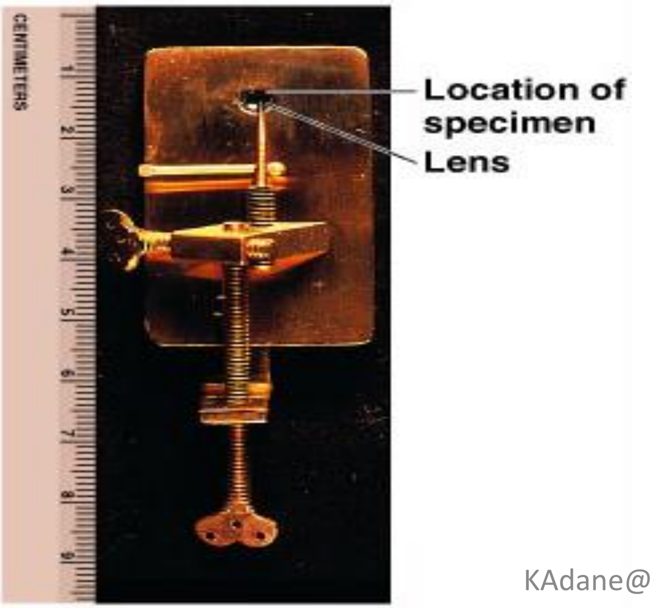
- Food spoilage

History of microbiology

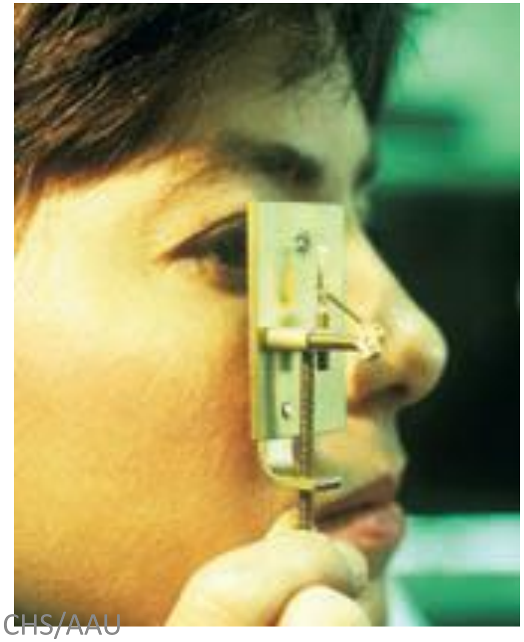
The First Observations

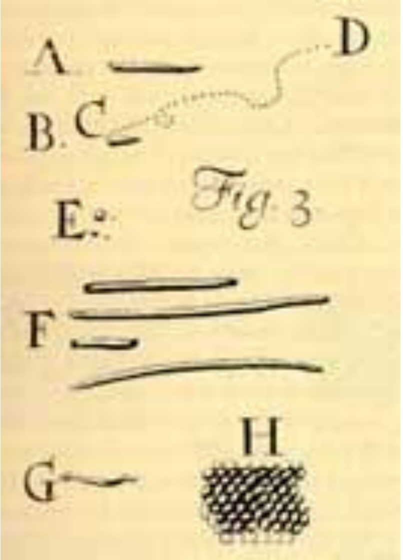
Antony van Leeuwenhoek (1632–1723)

- First person - invented microscope and discovered the microbial world.
- He saw minute moving objects which he called **“little animalcules”**



(b) Microscope replica





Spontaneous Generation vs. Biogenesis

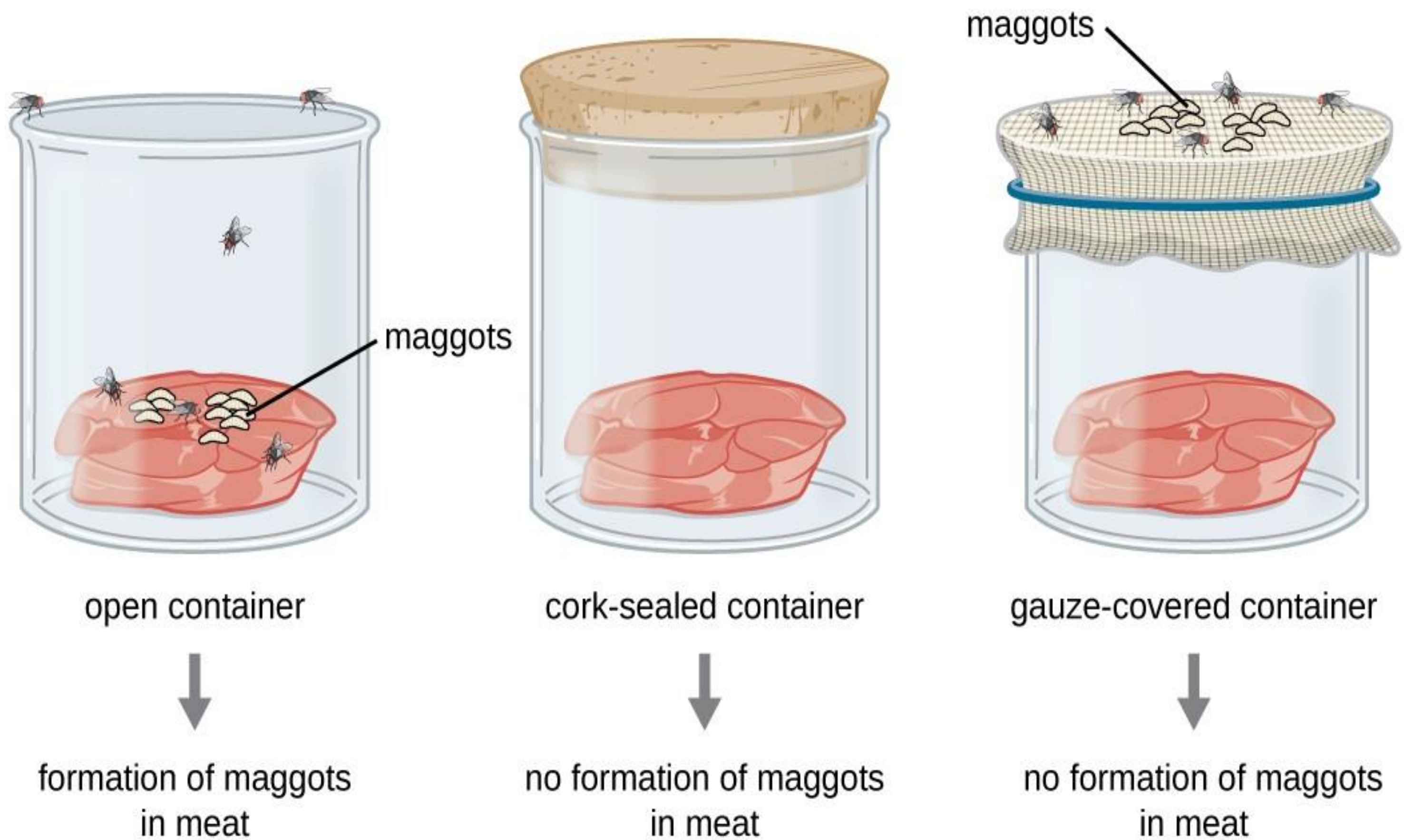
- **Spontaneous generation -** Living things arise from nonliving things
  - decaying meat → maggots and flies
  - grain → mice
- **Biogenesis -** Living cells can arise only from preexisting living matter.
  - e.g. animalcules arise from " germs" or "seeds" present in air

The Experiments

- In attempting to **prove or disprove** spontaneous generation, scientists performed controlled experiments.

Experiment #1

- 1668: **Francesco Redi** (biogenesis) placed decaying meat in different jars then made these observations:



Conclusion: maggots were the offspring of flies, not products of spontaneous generation

Experiment #2

- 1745: **John Needham** (spon gen) put boiled nutrient broth into flasks.

Conditions	Results
Nutrient broth placed flasks, boiled, and then sealed	Microbial growth
Conclusion: microbes arose spontaneously in broth from a "life force."	

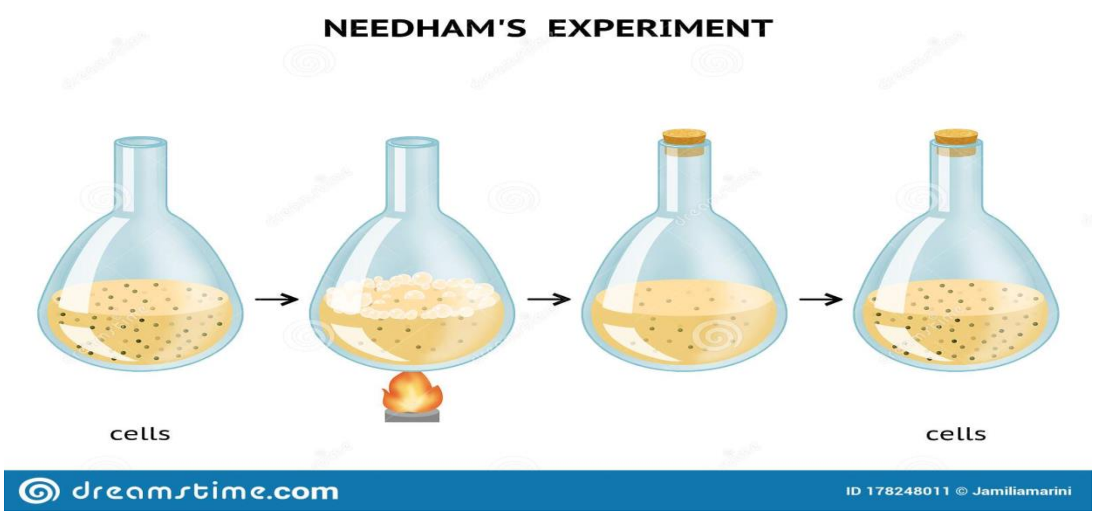
Experiment #3

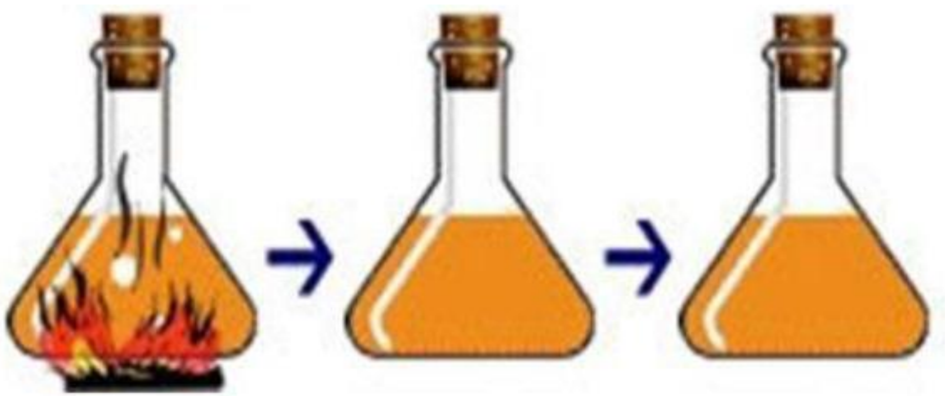
- 1767: **Lazzaro Spallanzani** (biogenesis) boiled nutrient solutions in sealed flasks.

Conditions	Results
Nutrient broth placed in flask, sealed, then boiled	No microbial growth
Suggested that microbes were introduced into these flasks from the air.	

Needham argued that life originates from a "life force" that was destroyed during Spallanzani's extended boiling.

Needham's and Spallanzani's Experiments



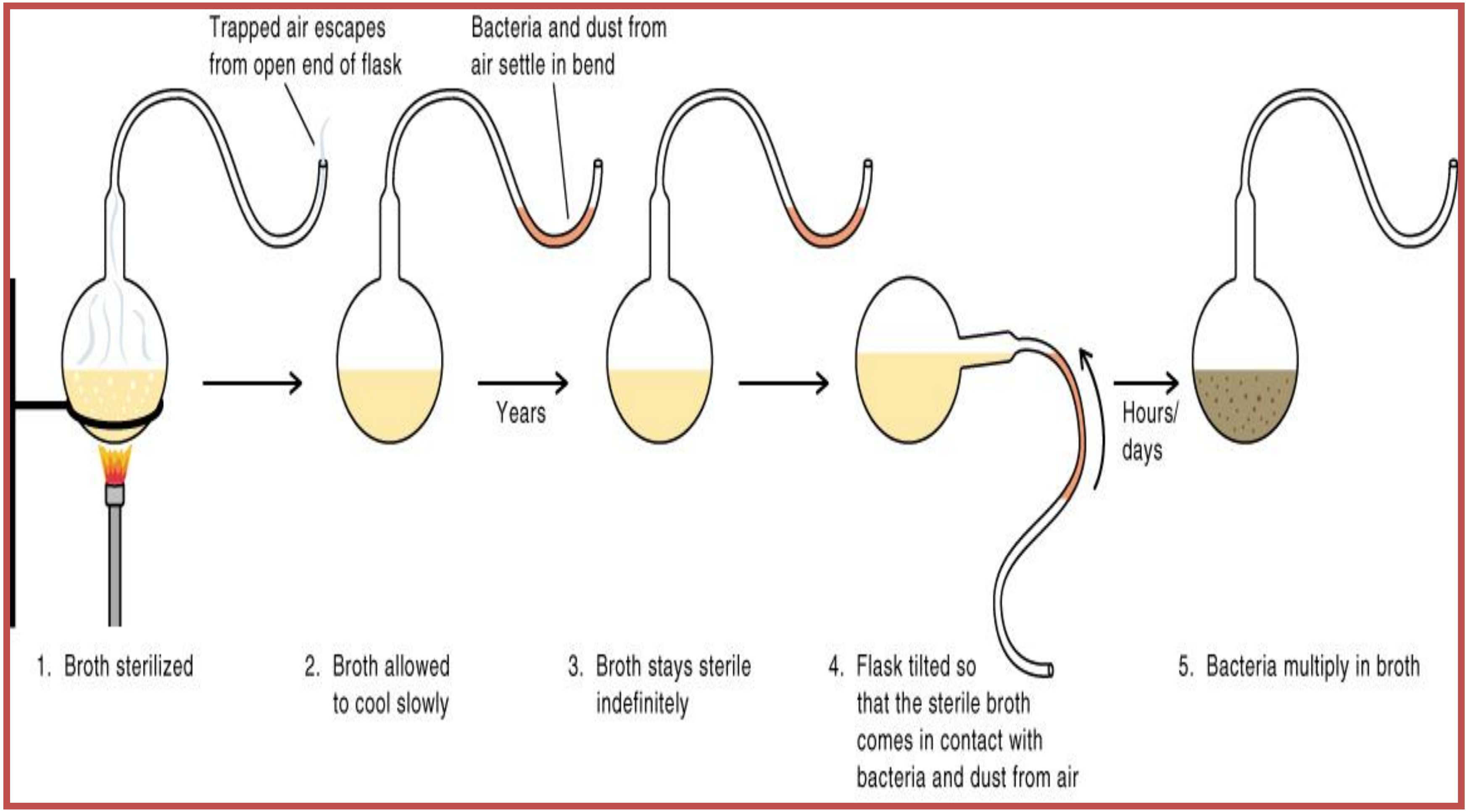


Experiment #4

- 1858**: Louis Pasteur** (biogenesis) demonstrated that microorganisms are present in the air
  - Filtered air through a cotton filter
  - Examined the cotton filter under a microscopic
  - Cotton was full of microorganisms

Experiment #5

- 1861: Pasteur's Swan-Necked Flasks experiment to further confirm his findings in the air



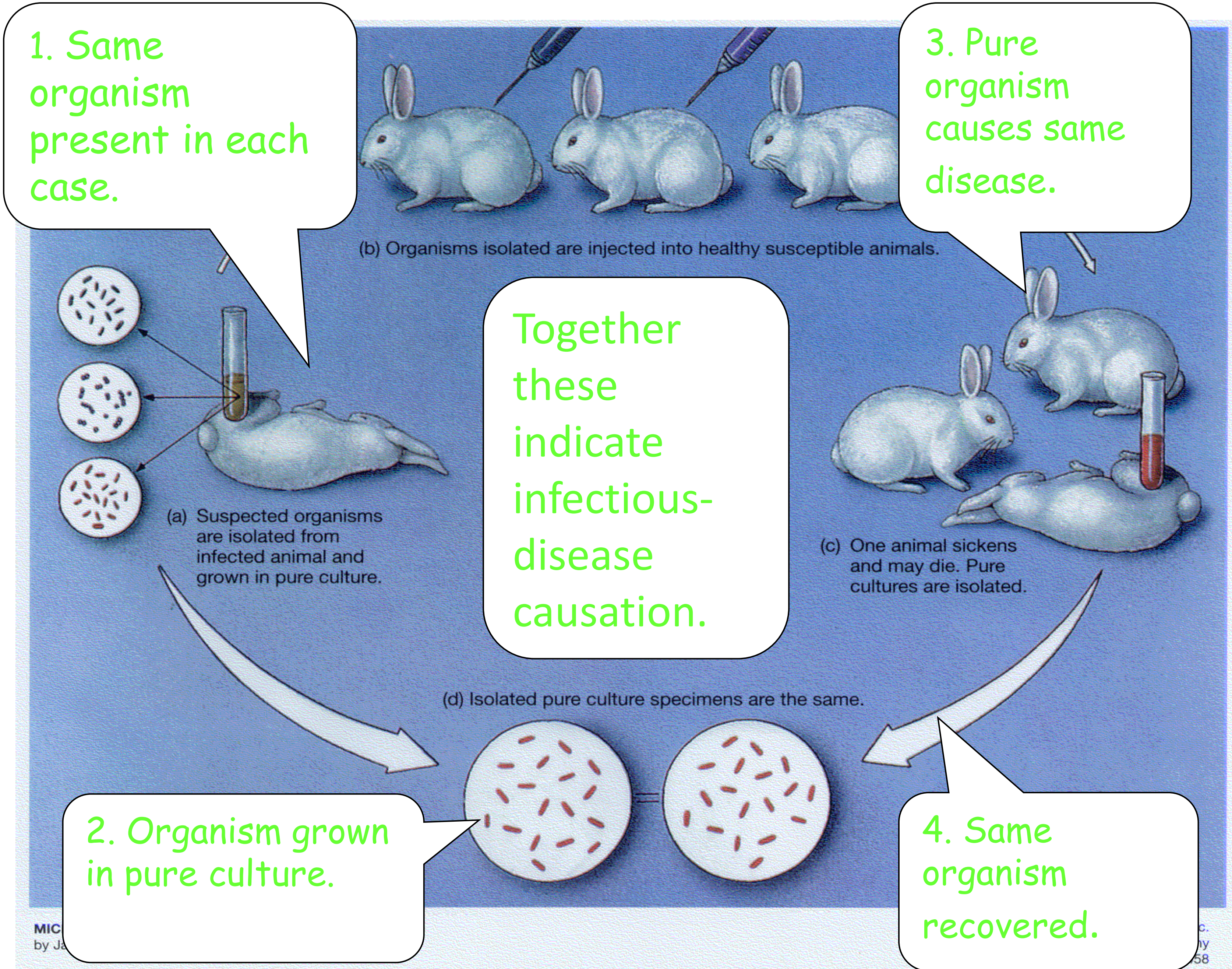
The germ theory of disease

- Certain diseases are caused by the invasion of the body by germs (microorganisms)
  - 1876- **Robert Koch-** established "scientific principles" to show a cause and effect relationship between a microbe & a disease
  - The principles are known as Koch's postulates

Koch's postulates

1. The same microbe is always associated with a specific disease.
2. The microbe must be isolated from the host with the disease and grown in pure culture
3. The specific disease must be reproduced when a pure culture of the bacteria is inoculated into a healthy susceptible host
4. The original microbe must be recovered from the experimentally infected host

Koch's Postulates



- Koch established the etiology of 3 diseases
  - 1. Cholera (fecal-oral disease)
    - *Vibrio cholerae*
  - 2. Tuberculosis (pulmonary infection)
    - *Mycobacterium tuberculosis*
  - 3. Anthrax (sheep and cattle)
    - *Bacillus anthracis*

Exceptions to Koch's Postulates

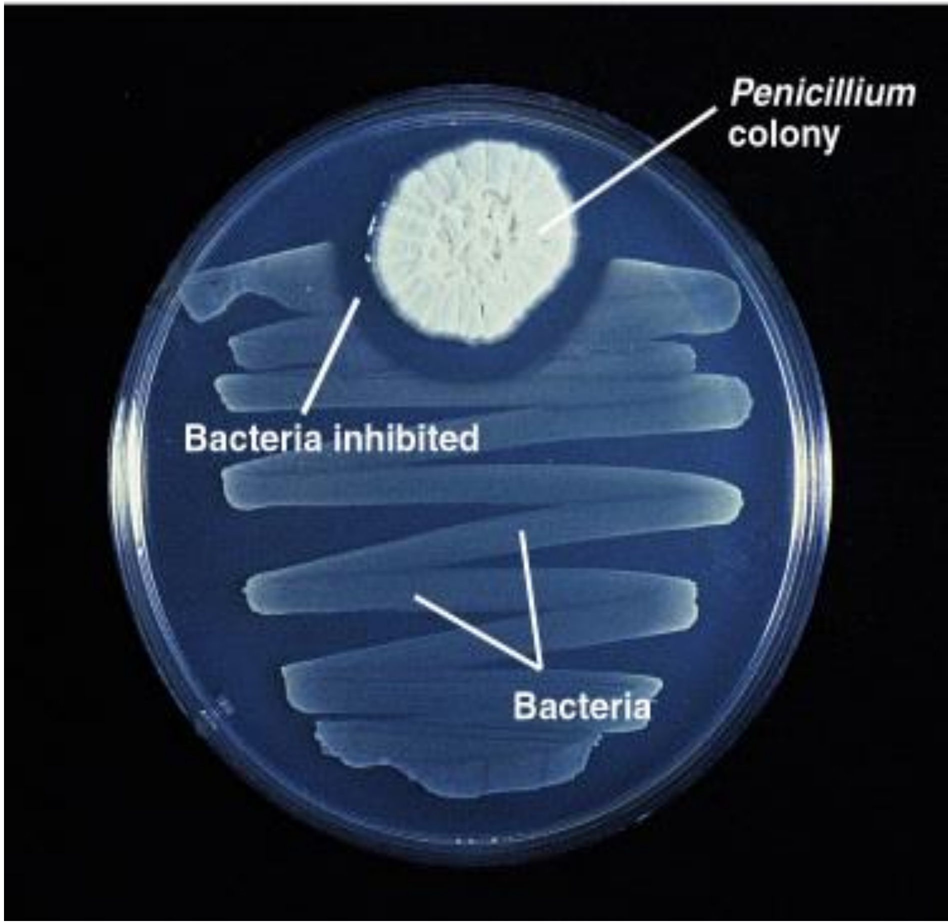
- Growth on artificial media is impossible in some cases
  - *Treponema pallidum* -Syphilis
  - *Mycobacterium leprae*-Leprosy
  - All viruses can not grow on artificial media
- Ethical objections- infecting subjects with infectious agents
  - No animal models for some disease
    - e.g. HIV/AIDS

Development of antimicrobial chemotherapy

- 1910: Paul Ehrlich developed Salvarsan (first synthetic chemotherapeutic agent) to treat syphilis

- 1929: Alexander Fleming, discovered penicillin(1st antibiotics)
- He observed that *Penicillium* fungus made an antibiotic, penicillin, that killed *S. aureus*.

-1942: Penicillin 1st Human Use



Taxonomy and classification of microorganisms

Taxonomy & classification of microorganisms

- **Taxonomy** is the science dealing with the description, identification, naming, and classification of organisms

- **Classification** is the ‘grouping’ of organisms based on particular characters

Taxonomy and classification of microorganisms

- Originally **2 Kingdoms**: Animals & Plants

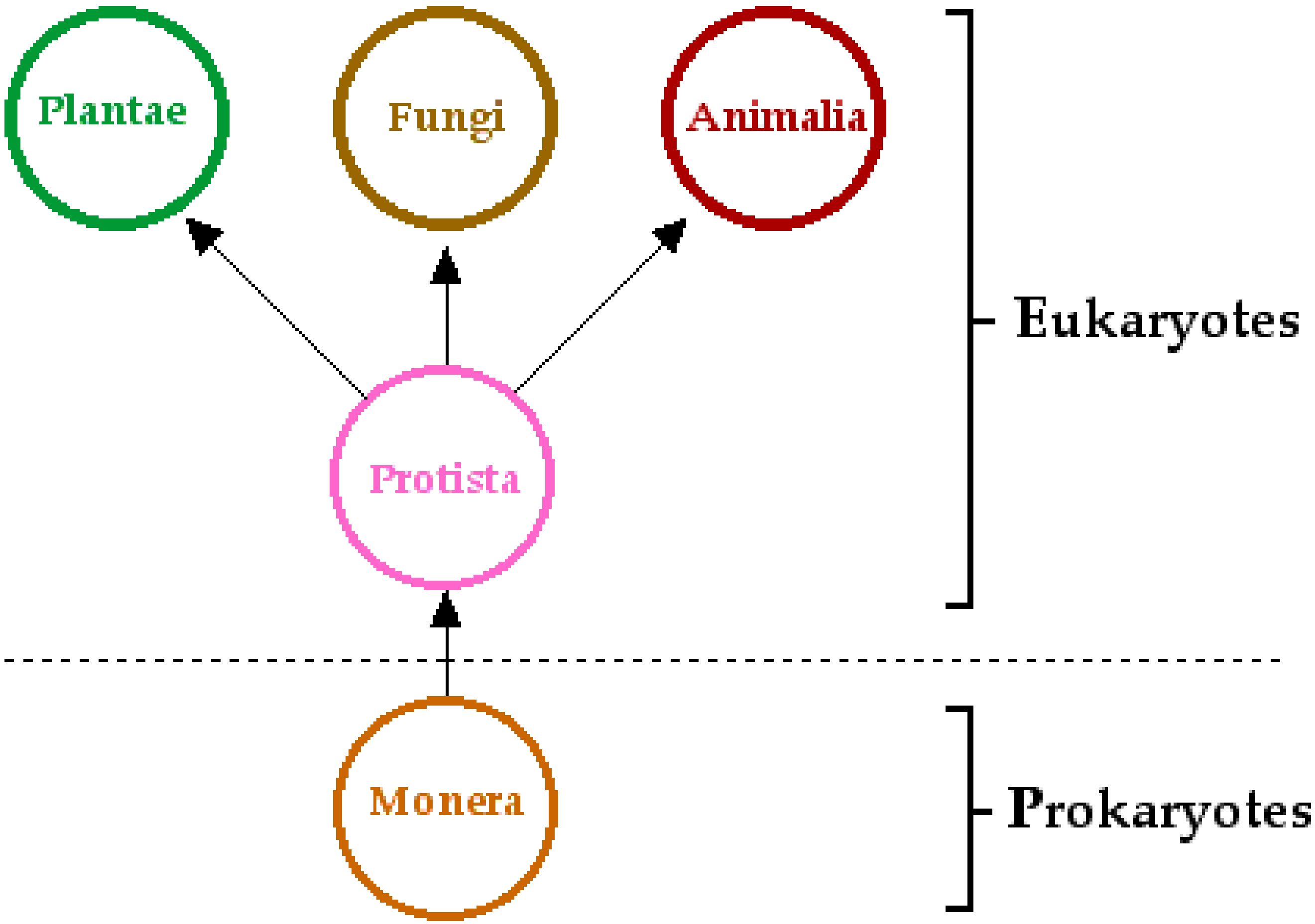
  (certain organisms do not fit)

- Robert Whittaker**: 5 Kingdoms** (1969)

  1. Prokaryotes/Monera(Bacteria & cyanobacteria)
  2. Protista
  3. Myceteae/Fungi
  4. Plant
  5. Animal

Taxonomy and classification of microorganisms

Whittaker's 5-kingdom system:



Binomial System of Nomenclature

- Binomial system was devised by Carolus Linnaeus
- Each organism has two names
  - Genus name – italicized and capitalized (e.g., Escherichia)
  - Species name– italicized but not capitalized (e.g., coli)

    Escherichia coli

  - Can be abbreviated after first use (e.g. *E. coli*)

Bacterial & fungal classification

Bacterial example

KINGDOM	Prokaryote/Monera
PHYLUM	Gracilicutes
CLASSS	cotobacteria
ORDER	Eubacteriales
FAMILY	Enterobacteriacea
GENUS	***Escherichia***
SPECIES	***coli***

Bacterial & fungal classification

Fungal example

KINGDOM	Eukaryote
PHYLUM	Ascomycota
CLASS	**Ascomycetes**
ORDER	Eurotiales
FAMILY	Trichocomaceae
GENUS	***Aspergillus***
SPECIES	***fumigatum***

Viral classification

International Committee on Taxonomy of Viruses (ICTV)

Host Organism; Particle Morphology & Genome Type

ORDER	-	suffix ***virales***
FAMILY	-	suffix ***viridae***
SUBFAMILY	-	suffix ***virinae***
GENUS	-	suffix **virus**
SPECIES	-	individual virus

Viral classification example

ORDER	-	*Mononegavirales*
FAMILY	-	*Filoviridae*
GENUS	-	Filovirus
SPECIES	-	Ebola virus

Eukaryotic vs. prokaryotic cells

- All organisms other than **viruses** and **prions** are made up of cells
- Bacteria are prokaryotes, all other organisms are eukaryotes

Prokaryotic cells

- A distinct nucleus is absent
- DNA is in the form of a single circular chromosome.
- Additional DNA is carried in plasmids
- Transcription and translation can be carried out simultaneously
- Ribosome size 70S

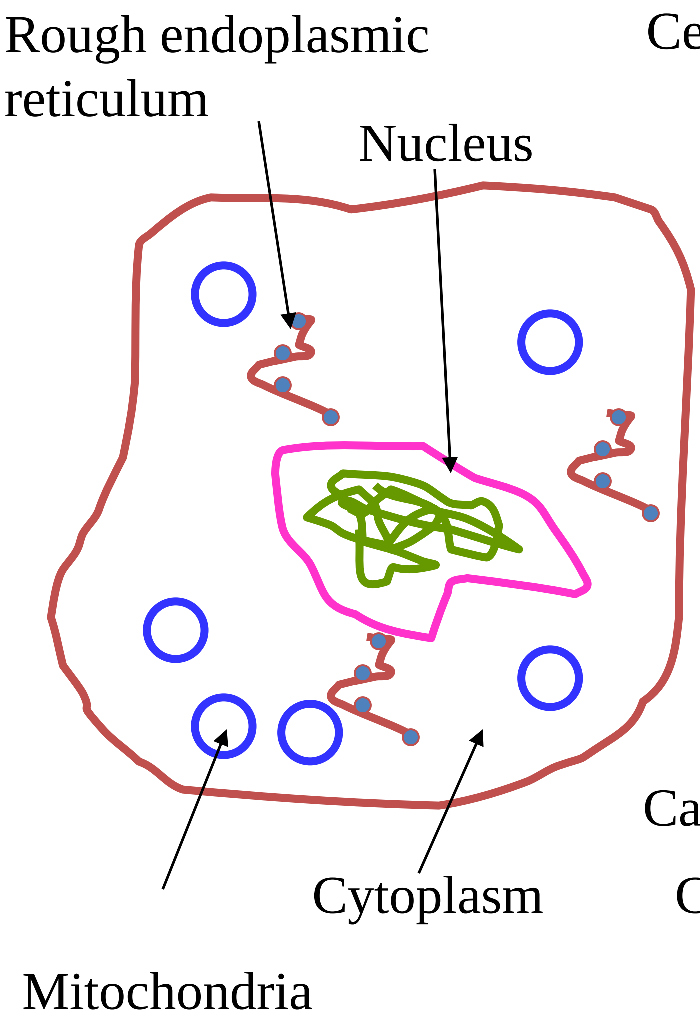
Eukaryotic vs. prokaryotic cells

Eukaryotes

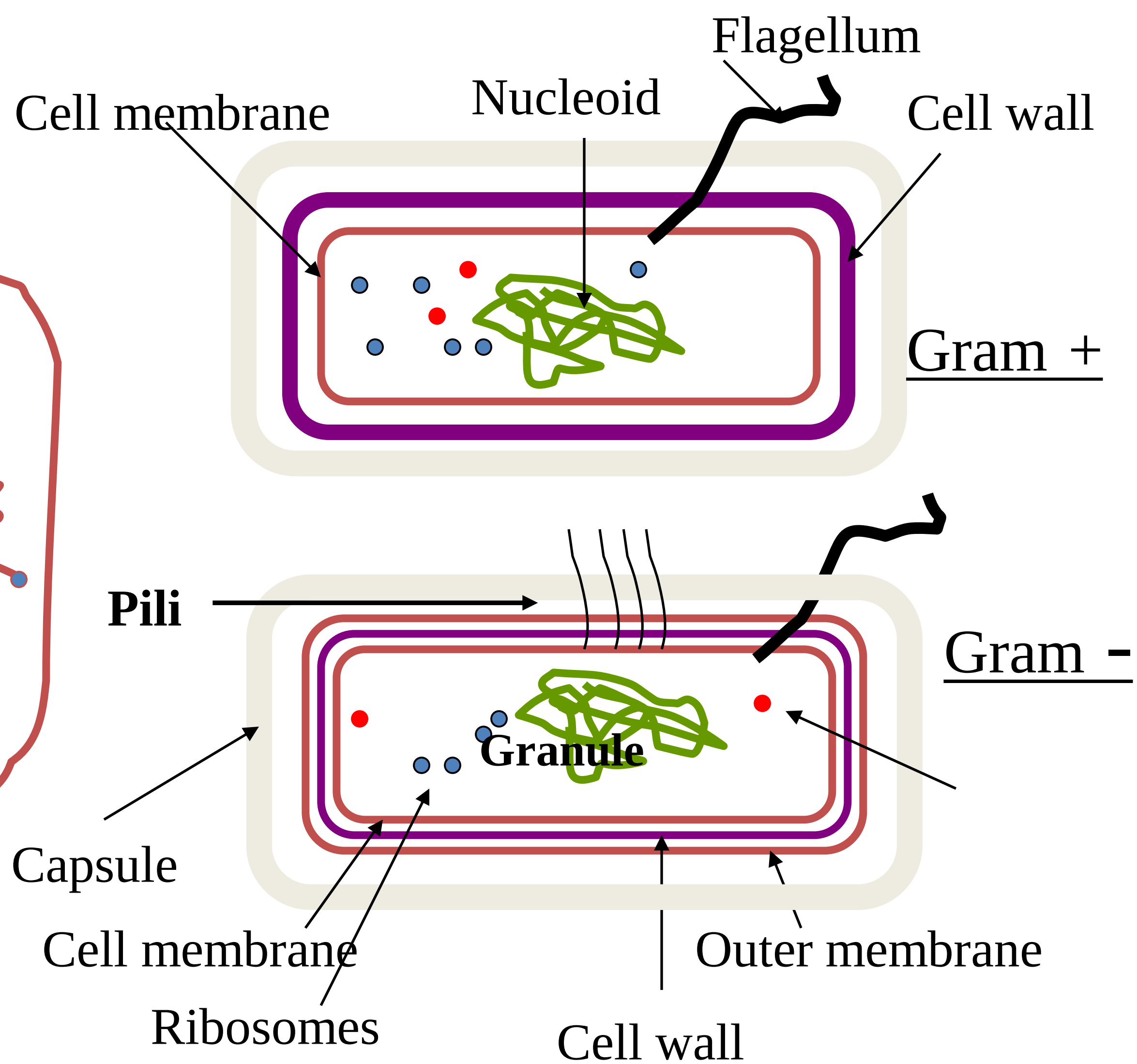
- Several chromosomes within a nucleus (linear)
- Nucleus bounded by a nuclear membrane
- Transcription requires formation of mRNA and movement of mRNA out of the nucleus into the cytoplasm
- Translation takes place on ribosomes
- Membrane-bound organelles
- Ribosome size 80S

Eukaryotic cell

(e.g. animal)



Prokaryotic cell



- **Bacterial Morphology and Growth**

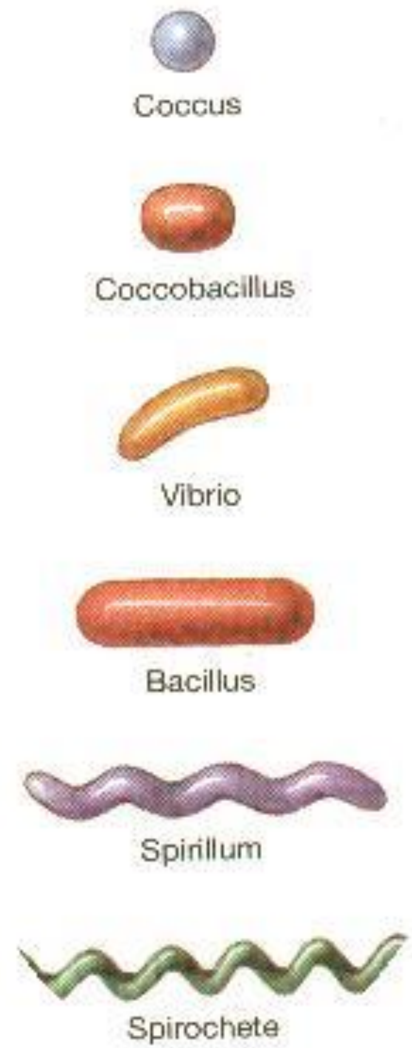
Outline of the lecture

- Bacterial size, shape and arrangements
- Basic bacterial structure
- Bacterial growth & factors affecting growth

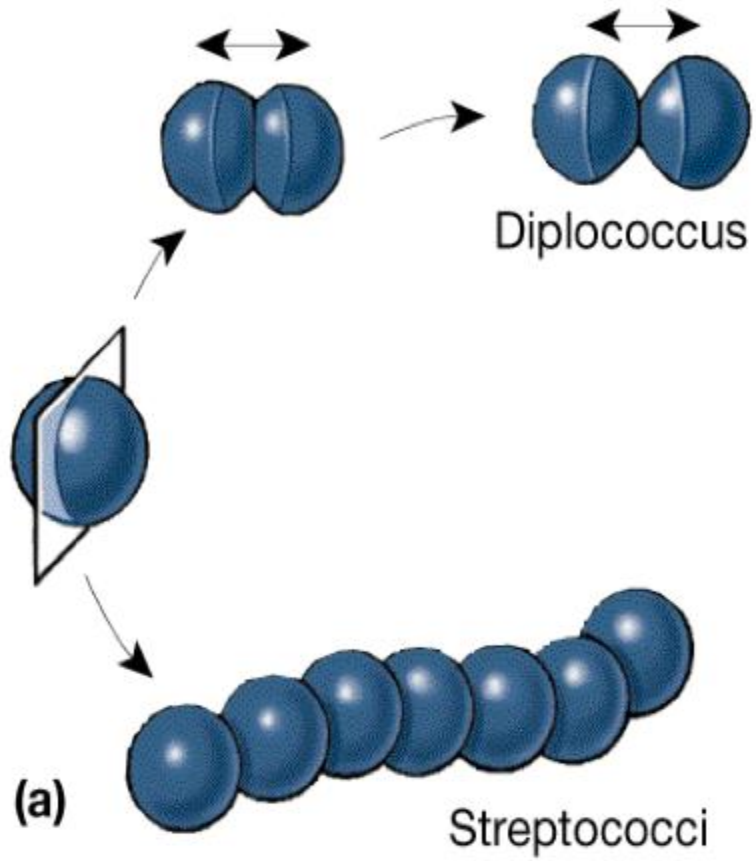
- **Bacterial size**
  - Bacteria range: 0.2 µm - 10 µm (typically 1 - 2 µm)
  - Mycoplasma sp. 0.15-0.3 µm in diameter (size of poxvirus)

Bacterial shapes

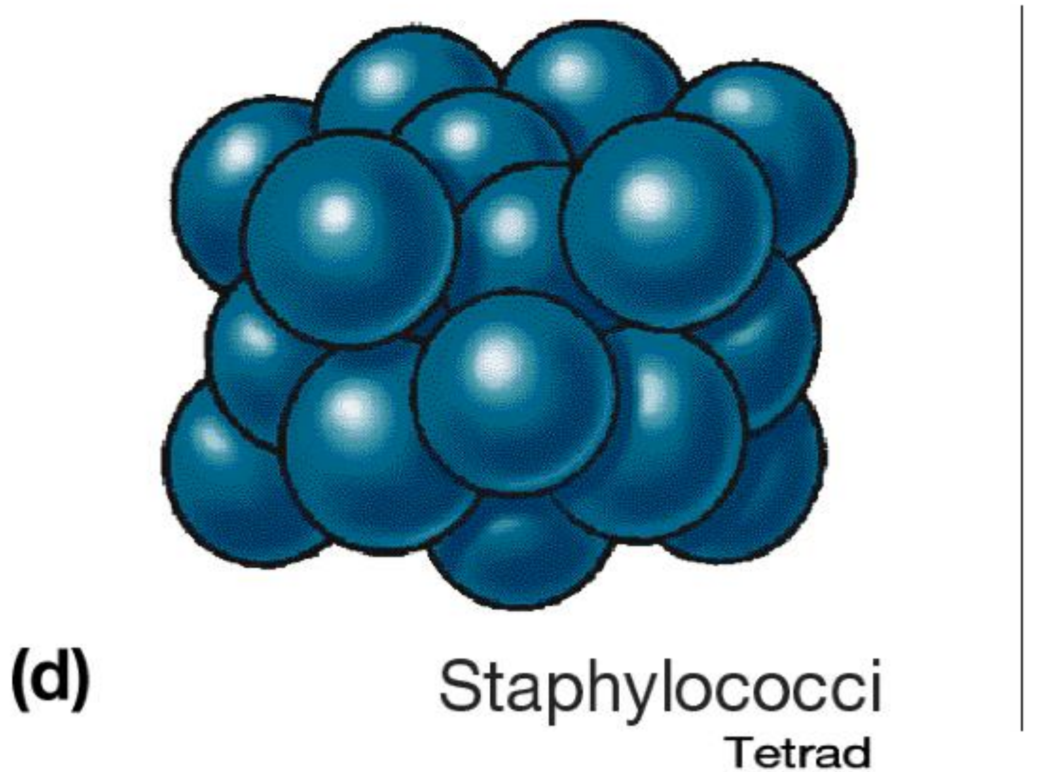
- **Coccus (pl = cocci)**: spherical cells
- **Coccobacillus**
- **Bacillus (pl = bacilli);** rod-shaped cells
- **Vibrios** - single curve
- **Spirillum (pl spirilla):** rigid, two or more curves
- **Spirochete**: flexible, two or more curves, wavy
- **Pleomorphic** - display different shapes
  - Mycoplasma pneumonia



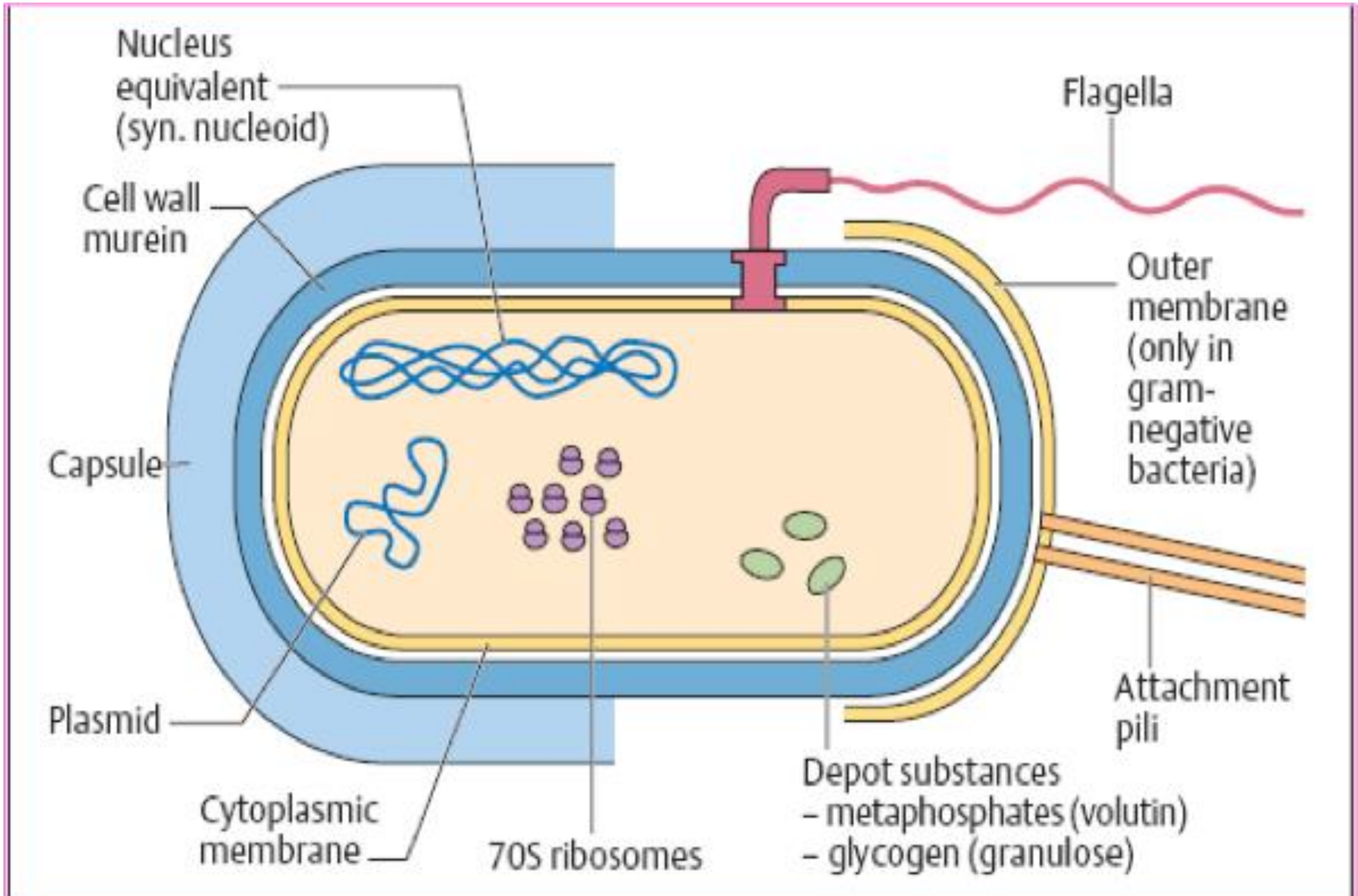
Bacterial cell arrangements



Clusters of geometrically arranged cocci



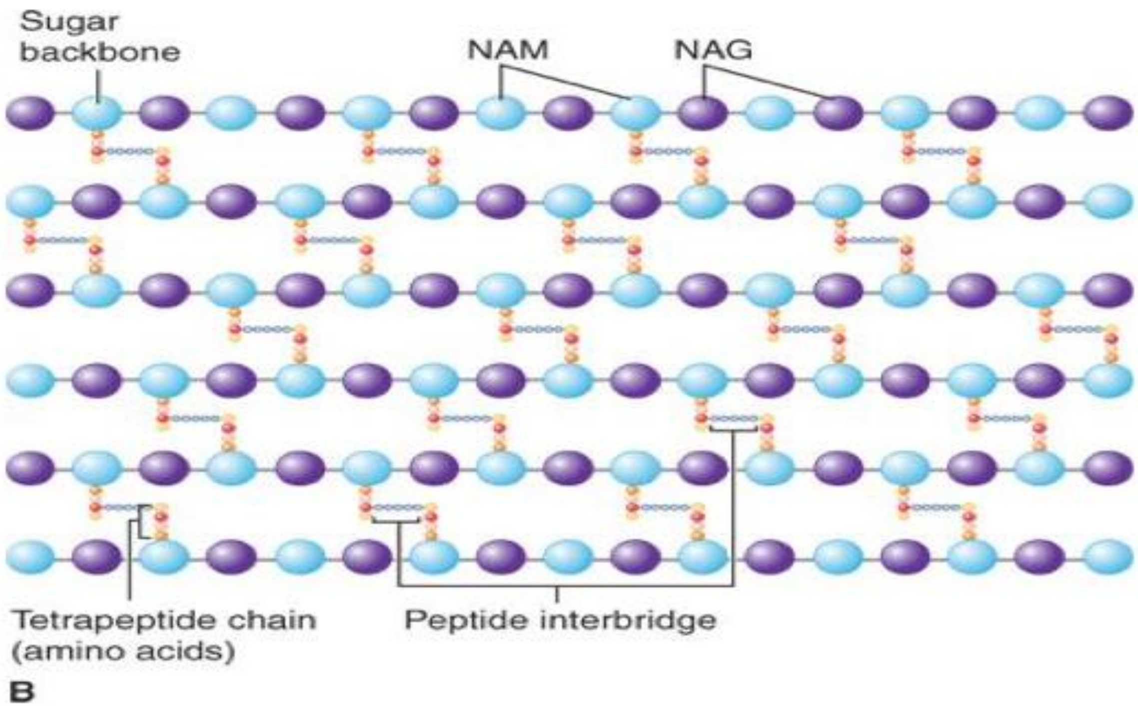
Basic Bacterial Cell Structure



Composition of bacterial cell wall

- Cell wall composed of peptidoglycan (Murein) layer
- **Consists of three parts:**
  - Sugar backbones (Glycans)
  - Peptide side chains
  - Peptide cross-bridges

Peptidoglycan layer

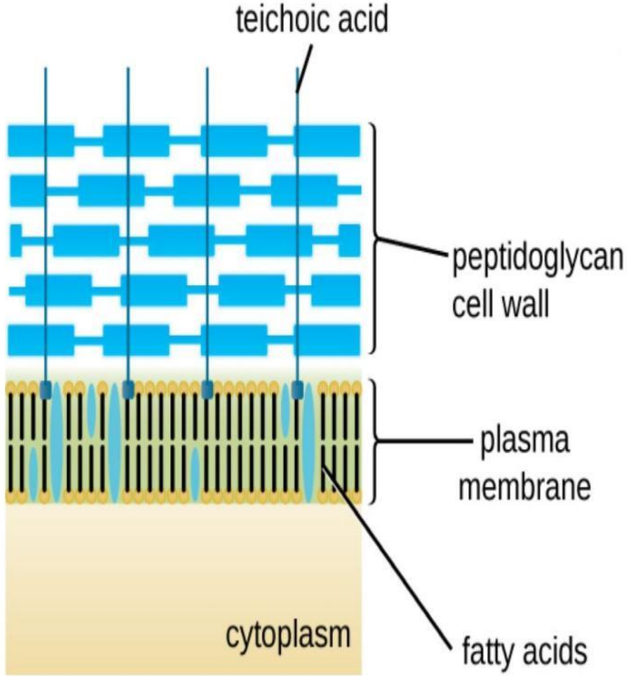


Source: Warren Levinson: Review of Medical Microbiology and Immunology, 14th Edition, www.accessmedicine.com

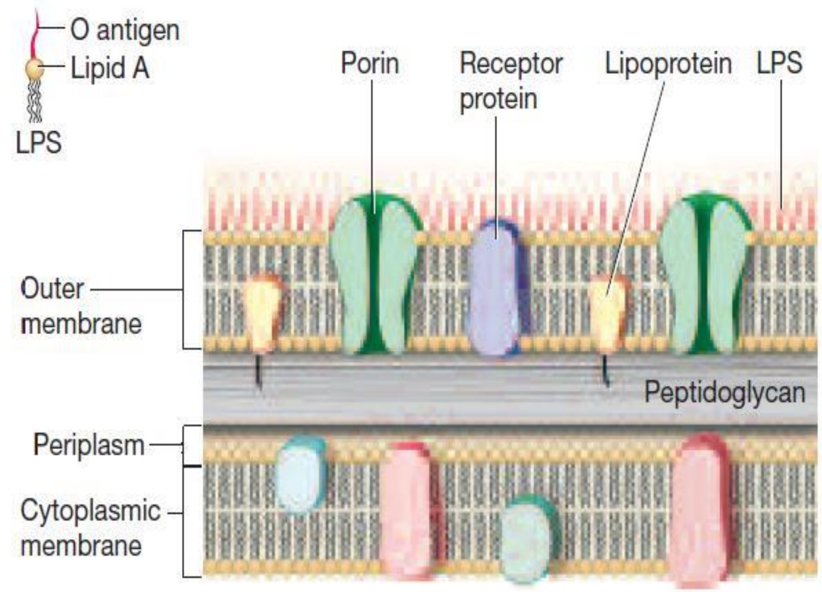
Gram +ve vs. Gram-ve bacteria

- **Gram +ve bacterial cell wall**
  - Thick peptidoglycan
  - Teichoic acid
- **Gram-ve bacterial cell wall**
  - Outer membrane with lipopolysaccharide (LPS)
  - Thin peptidoglycan

Bacterial Cell Wall

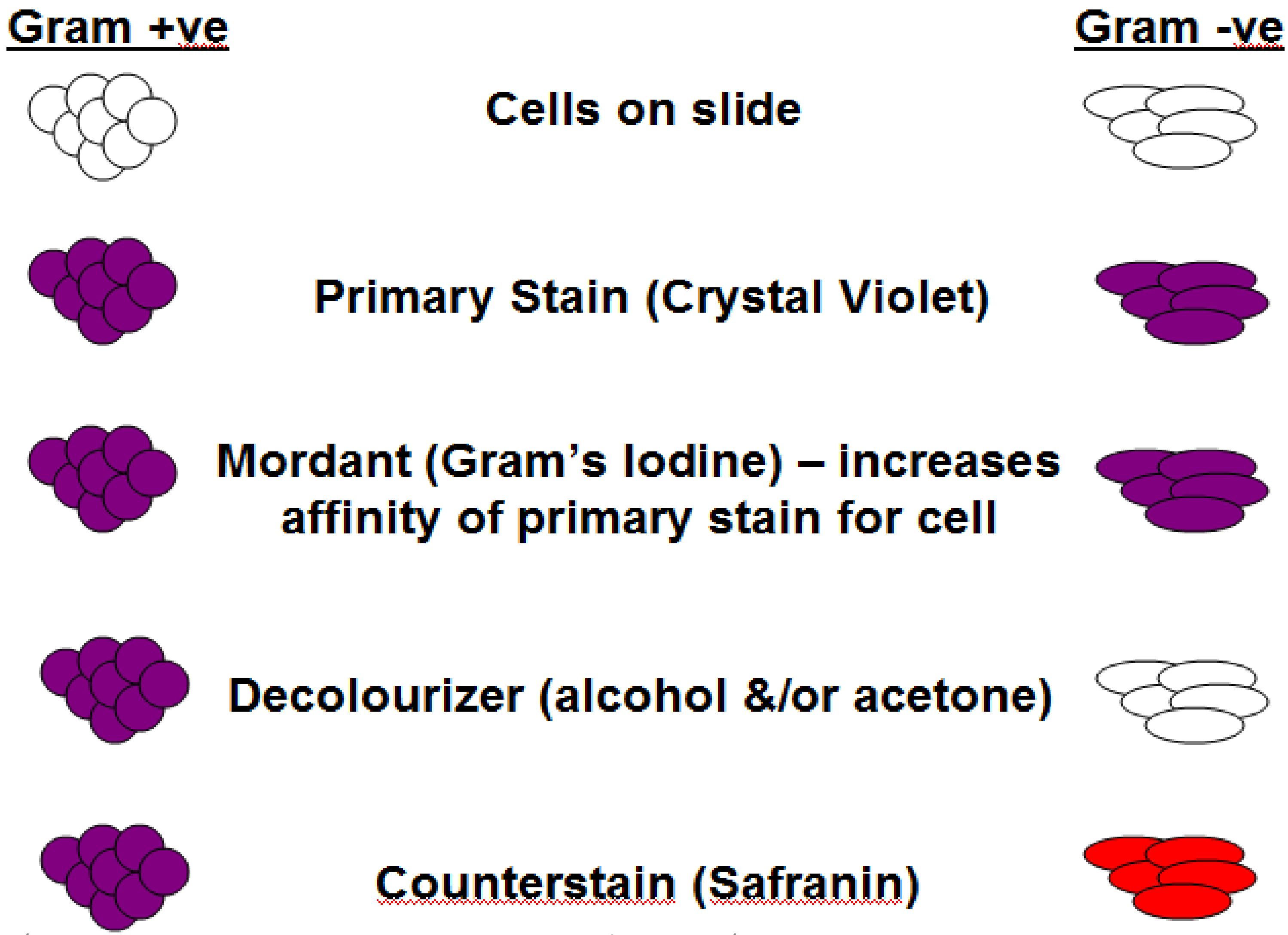


gram-positive bacteria



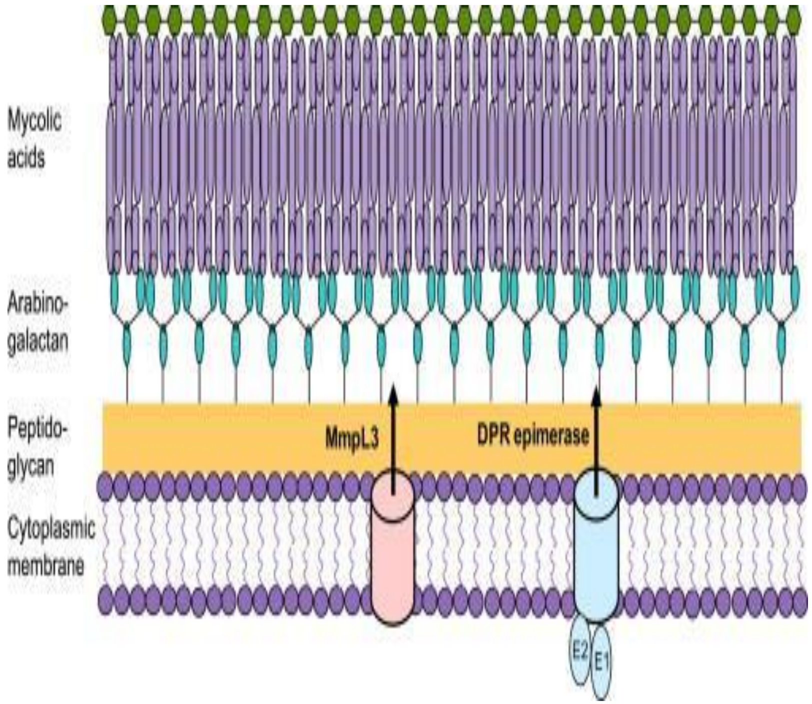
Cell Wall of Gram Negative Organism

Gram +ve vs. Gram-ve bacteria



Bacteria with Atypical Cell Walls

- **Mycobacterium & Nocardia**
  - Lipid rich cell wall
  - Contain waxes/mycolic acids
  - Can not be stained by gram staining
  - Use Acid-fast staining/ Ziehl-Neelsen staining



Cell wall-lacking bacteria

- **The Mycoplasmas**
  - Containing no peptidoglycan
  - Inherently resistant to cell-wall acting antimicrobial agents

Spores

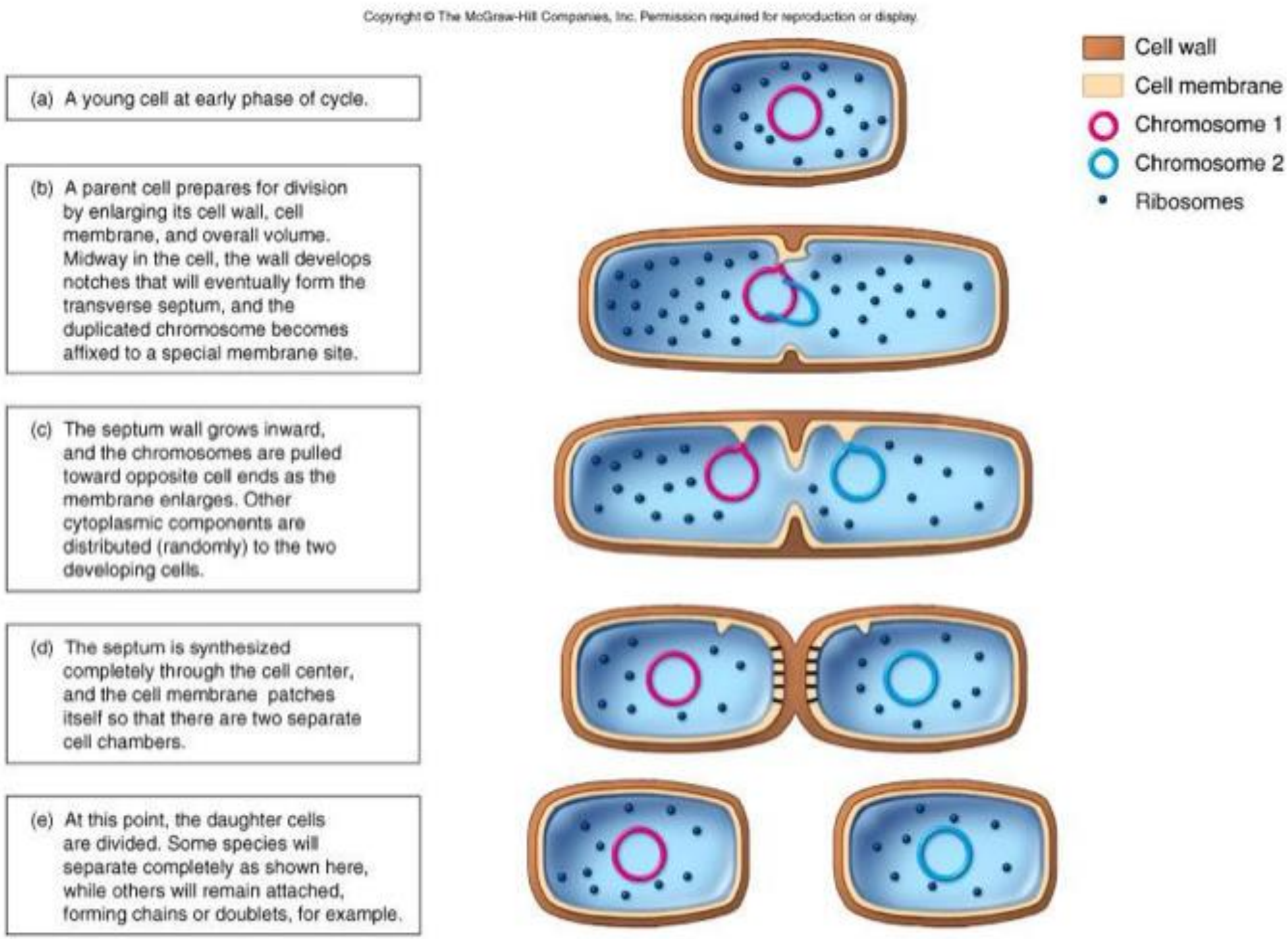
- Specialized structures produced in environmental stress
  - Bacillus & Clostridium spp. produce spores
- Resistant structure
  - Heat, irradiation, cold, chemical disinfection & drying
  - Boiling for hours may not kill them
- Bacillus anthracis spores
  - Used in biological warfare

- **Bacterial growth**

Growth of bacteria

- Increase in number of cells, not cell size
- One cell becomes colony of millions of cells

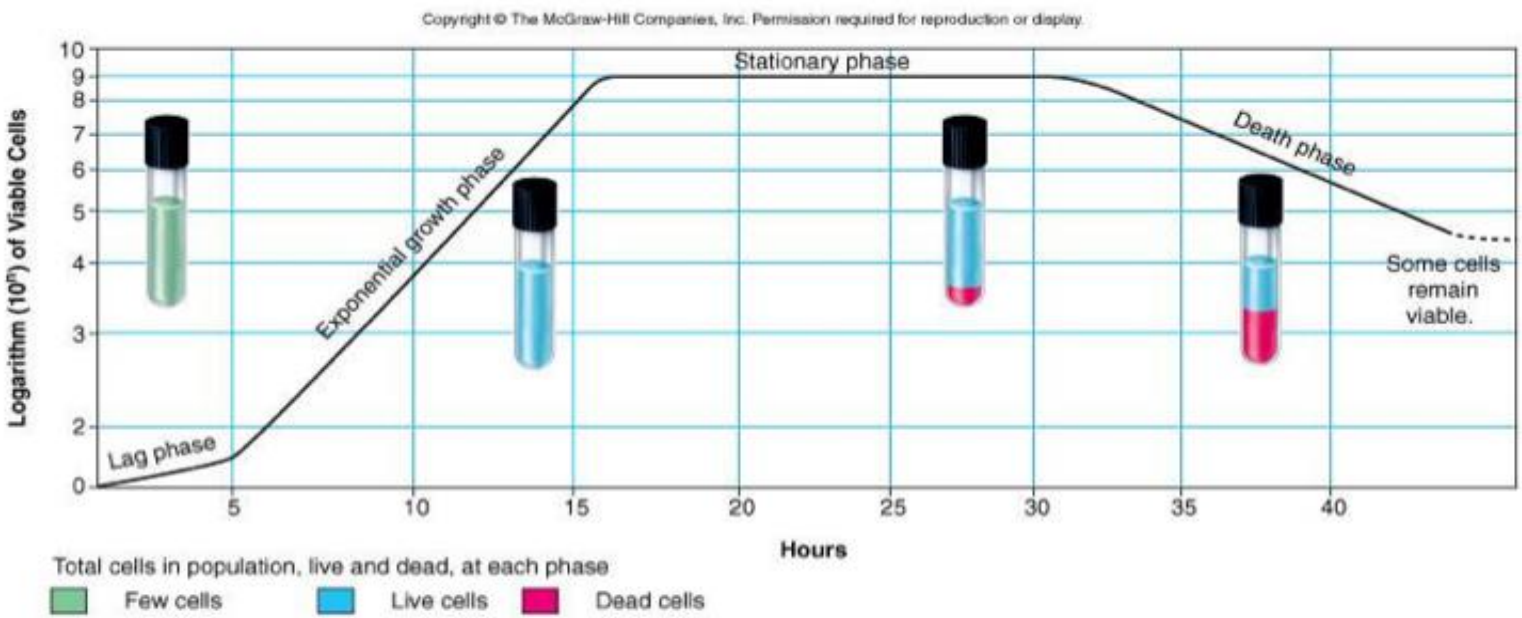
Bacterial Division: binary fission



Generation time

- Time required for cell to divide/for population to double
- Average for bacteria is 1-3 hours
  - *E. coli* generation time = 20 min
    - 20 generations (7 hours), 1 cell becomes 1 million cells
  - *Mycobacterium tuberculosis* requires 360 mins- slow grower

Standard Growth Curve



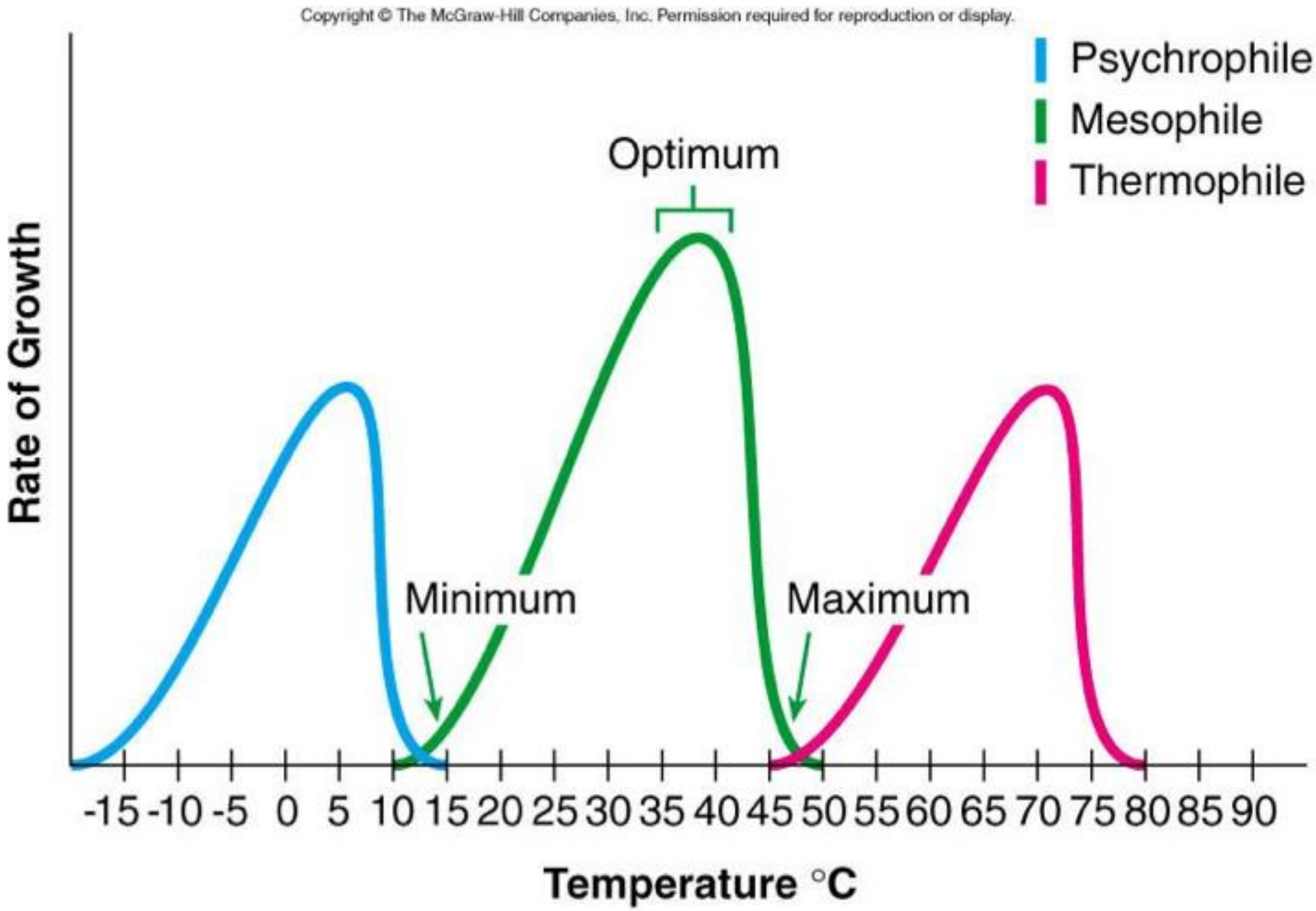
Factors Regulating Growth

- **Physical**
  - Temperature
  - O_2
  - pH
- **Chemical**
  - Minerals, and organic growth factors
  - Sources of carbon and nitrogen

Physical factors

Temperature Optima

- **Psychrophiles**: cold-loving
  - Contaminants of food/dairy products
- **Mesophiles**: moderate temperature-loving
  - Most human pathogens
- **Thermophiles:** heat-loving
  - Not involved in human infections
- Each has a minimum, optimum, and maximum growth temperature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Psychrophile
Mesophile
Thermophile
Optimum
Minimum
Maximum
Rate of Growth
-15 -10 -5 0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90
Temperature °C

Physical factors

- **Oxygen Requirements**
  - Obligate aerobes – require O_2
  - Facultative anaerobes – can use O_2 but also grow without it
  - Obligate anaerobes – die in the presence of O_2

Physical factors

pH requirements

- Neutralophiles
  - Most organisms grow best at a pH of 6.0–8.0
- Acidophiles
  - Some forms have optima as low as pH 3.0
- Alkaliphiles
  - Others have optima as high as pH 10.5

Chemical factors

Nutrient Requirements

- **CHNOPS** - Macronutrients
  - Carbohydrates, Lipids, Protein & Nucleic acids
- **Cations ; Minerals** : K^+, Ca^{2+}, Mg^{2+}, Fe^{2+}/Fe^{3+}
  - Potassium - enzyme activity
  - Calcium - heat resistance
  - Magnesium - enzyme cofactor
  - Iron - cytochromes, cofactor
- **Trace Elements** (Mn, Zn, Co, Ni, Cu, Mo)
  - Enzymes or cofactors

Chemical factors

- **Carbon sources**
  - CO_2 = autotroph
  - Organic = heterotroph
- **Energy sources**
  - Sunlight = phototroph
  - Organic = chemotroph

- **Thank you!**